Le-Chris Wang

Curriculum Vitae

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Education

Aug. 2025 – May 2030 **Princeton University**, *Princeton, NJ, USA*
PhD in Astrophysical Sciences

Aug. 2021 – May 2025 **Johns Hopkins University**, *Baltimore, MD, USA*
BSc in Computer Science and Physics, Both with Departmental Honors
Minor: Applied Mathematics & Statistics and Pure Mathematics
Advisor: Prof. Kevin Schlaufman, Prof. David Sing

Publications

First-authored/equivalent Publications

7. **Wang, L. C.**, Sagnick Mukherjee, Stephen P. Schmidt et al., "Asymmetric Aerosol Distribution on the Terminators of the Warm Saturn WASP-69 b Revealed by JWST NIRISS/SOSS" Submitted to *ApJL*.
6. **Wang, L. C.** & Schlaufman, K. C., "Elemental Abundance Trends with Condensation Temperature are Unrelated to Planet Formation" Submitted to *ApJ*.
5. **Wang, L.-C.** & Winn, J. N., How Many Transiting Giant Planets Can JWST Search for Moons and Rotational Oblateness?" *ApJL* (2026).
4. **Wang, L.-C.**, Sing, D. K., & Rustamkulov, Z., "A Comprehensive Analysis of the Panchromatic Transmission Spectrum of the Hot-Saturn WASP-96 b: Nondetection of Haze, Possible Sodium Limb Asymmetry, Stellar Characterization, and Formation History" *AJ* (2026).
3. Liu, R.*, **Wang, L.-C.***, Rustamkulov, Z., & Sing, D. K., "Unveiling the atmosphere of the super-Jupiter HAT-P-14b with JWST NIRISS and NIRSpec" *AJ* (2025, *: Co-first author).
2. Gou, X.*, Pan, X.*, **Wang, L.-C.***, "General Relativity Testing in Exoplanetary Systems " *IOP Conf. Ser.: Earth Environ. Sci.* (2021). (*: Equal contributions).
1. Zheng, Y., Wang, X., **Wang, L.-C.*** et al., "Test of Bell's and Mermin's inequalities on Quantum Computer" *2020 2nd International Conference on Information Technology and Computer Application* (2020) (*: Corresponding Author).

Coauthoed Publications

8. Katherine A. Bennett et al., incl. **Wang, L.-C.**,"A JWST NIRISS/SOSS Transmission Spectrum of the Exo-Neptune HAT-P-11 b Reveals Aerosols and a High Metallicity In Agreement with the Mass-Metallicity Trend", submitted to *AAS Journals*.
7. Ashtari, R. et al., incl. **Wang, L.-C.**,"A Clearer View of HAT-P-1 b: JWST NIRSpec G395H Reveals a Cloud-free Atmosphere of Water, Carbon Dioxide, and Possibly Hydrogen Sulfide", submitted to *AAS Journals*.
6. Sing, D. K. et al., incl. **Wang, L.-C.**,"A JWST transiting survey of FGK stellar limb darkening: empirical evidence for quadratic laws and atmospheric model comparisons", submitted to *AAS Journals*.
5. Schmidt, S. P. et al., incl. **Wang, L.-C.**,"Mitigating Charge Migration in JWST NIRISS Reveals That KELT-7 b is a Metal-enriched Ultra-hot Jupiter Orbiting a Young Metal-rich Star" preprint, submitted to *AAS Journals*.

4. Mukherjee, S. et al., incl. **Wang, L.-C.**,"Cloudy mornings and clear evenings on a giant extrasolar world" *Science* (2026).

3. Schmidt, S., MacDonald, R., Tsai, S., Radica, M., **Wang, L.-C.** et al., "A Comprehensive Reanalysis of K2-18 b's JWST NIRISS+NIRSpec Transmission Spectrum" *AJ* (2025).

2. Wang, G., incl. **Wang, L.-C.**, "A Revised Density Estimate for the Largest Planet, HAT-P-67 b" *AJ* (2025).

1. Chen, H., Clement, M. S., **Wang, L.-C.**, & Gu, J. T., "Born Dry or Born Wet? A Palette of Water Growth Histories in TRAPPIST-1 Analogs and Compact Planetary Systems" *ApJL* (2025).

Awards & fellowships

2025 **First-Year Fellowship in the Natural Sciences and Engineering**
One-year awards for doctoral students newly admitted by a degree program in Natural Sciences or Engineering for the first-year of study.

2024 **IDIES Summer Student Fellowship**
$6,000, JHU, competitive research fellowship awarded to 5 undergraduates conducting data-intensive research.

2024 **ΣΠΣ Physics Honor Society**
Awarded for outstanding academic achievement in physics and astronomy.

2023 **Summer Provost's Undergraduate Research Award**
$6,000, JHU, competitive research fellowship (36 out of 400+ applicants).

2023 **Dean's ASPIRE Grants**
$2,474, JHU, competitive research fellowship; awarded to 10 undergraduates per year.

2022 **Hophacks 2nd place**
Hopkins's premier 36-hour hackathon. 2/40. $512 prize.

2022 **Quest2Learn Most Innovative Platform to Help with Learning**
Awarded for creating an application that helps with learning.

2022 **Bloomberg Distinguished Professor Fellowship**
$6,000, JHU, competitive research fellowship; awarded to 2 physics undergraduates.

2021–2025 **Dean's List**
Excellence in academics. Awarded every semester (8/8).

Talks & Presentations

Dec. 2025 **Unveiling the Formation History of the Cloud-free Exoplanet**
International Conference on Exoplanets and Planet Formation

Oct. 2024 **Planet Formation with N-body Simulations**
2024 IDIES Annual Symposium

April 2024 **FIREFLy-SOSS: Exoplanet Transit Light Curves Extraction Pipeline for JWST NIRISS-SOSS Observations**
Departmental Undergraduate Research Showcase, Johns Hopkins University, MD

April 2024 **Is The Formation Of Planets The Cause of Solar Atypical Abundance Pattern?**
Johns Hopkins University DREAMS Symposium

April 2024 **Characterization of Cloud-free Hot-Saturn WASP-96b with Joint JWST, Hubble, VLT, and Spitzer Transmission Spectroscopy**
Johns Hopkins University DREAMS Symposium

Jan. 2024 **Elemental Abundance Trends with Condensation Temperature are Unrelated to Planet Formation**
243rd Meeting of the American Astronomical Society, New Orleans, LA

June 2023 **Elemental Abundance Trends with Condensation Temperature are Unrelated to Planet Formation**

	Origins of Solar Systems Gordon Research Conference, Mount Holyoke College, MA
June 2023	**Stellar Elemental Abundance Patterns: Implications for Planet Formation**
	No-PhD Journal Club, Johns Hopkins University, MD
Aug. 2022	**Optimizing JWST BOTS Transit Light Curve Fitting**
	The Center for Astrophysics Research Experience, Johns Hopkins University, MD

Telescope Allocations

2026 Q3	**Apache Point Observatory, ARCTIC, 2 nights** *Star Monitoring and Refining Ephemerides for Exoplanets Favorable for Oblateness and Exomoons* PIs: **Wang, L.-C.** & Wang, G.
2024 Q4	**Apache Point Observatory, ARCSAT, 14 nights** *Complementary Flare Monitoring of the Enigmatic Triple Red Dwarf LTT 1445ABC* PIs: Rustamkulov, Z., Bennett, K., CoIs incl. **Wang, L. C.**
2024 Q3	**Apache Point Observatory, ARCTIC, 3 nights** *Synergistic Cool Star Monitoring: Characterization of Starspots* PIs: Rustamkulov, Z., Allen, N., **Wang, L. C.**, Wang, G.

Teaching appointments

2024 Spring	Teaching Assistant, AS.171.108 General Physics II (Undergraduate, 23 students)
2023 Fall	Teaching Assistant, AS.171.107 General Physics I (Undergraduate, 46 students)
2023 Spring	Teaching Assistant, AS.171.101 General Physics I (Undergraduate, 46 students)
2022 Fall	Teaching Assistant, AS.171.101 General Physics I (Undergraduate, 23 students)

Skills

Programming	Python, C/C++, Java, Assembly, Fortran, Matlab, R, HTML, CSS, JavaScript, Bash, rust, go, React, flask, Node.js
Data science	sql, MySQL, PostgreSQL, Pytorch, TensorFlow, OpenCV, docker
Astronomy Softwares	DS9, Siril, MESA (stellar structure), Rebound (N-body), Mercury (N-body), petitRADTRANS (atmospheric retrieval)
Observation Experience	The Morris W. Offit Telescope (half-meter telescope at JHU), ARC 3.5m telescope at Apache Point Observatory
Languages	English, Chinese, French
Other	LaTeX, Git, Slurm, Mathematica, JupyterLab, Adobe Lightroom, Adobe Photoshop, Blender, A Cappella, Marathon